

Multivariate Models & Interactions

GOVT 10: Quantitative Political Analysis

This chapter extends regression beyond a single predictor. We add control variables to hold confounders constant, interpret coefficients in a multivariate model, and then introduce categorical predictors and interaction terms for relationships that depend on context.

1 Multivariate Models

Why This Matters

Simple regression tells you that two variables move together. It does not tell you why. Countries with high foreign aid tend to have worse governance; that does not mean aid causes bad governance — it may be that bad governance attracts aid. Districts with longer terms in office tend to have less accountability; that does not mean term length causes unaccountability — it may be that uncompetitive districts both elect long-term incumbents and produce weaker accountability. Multivariate regression lets us hold these competing explanations constant and ask: does the relationship survive?

1.1 The Problem with Simple Relationships

A simple regression of Y on X gives an honest summary of how the two variables move together in your data. It does not tell you whether X actually drives Y . That is because almost every interesting political relationship is tangled up with other variables that influence both sides at once.

Consider the well-known observation that countries that trade more with each other fight less. Read at face value, this looks like evidence that trade prevents war. But trading partners also tend to be geographically close, economically developed, democratic, and often formally allied. Any of these could be doing the real work. Trade and peace may both be consequences of a deeper variable like alliance membership, with no causal link between them at all.

This is the basic problem of confounding: a variable Z that affects both X and Y will produce an apparent X – Y relationship even when X has no effect on Y whatsoever. Simple regression cannot tell the difference. Multivariate regression is our first tool for cutting through these tangled relationships.

Confounders are everywhere

A confounder is a variable Z that causes both X and Y . Ignoring Z makes the simple X – Y relationship look stronger, weaker, or even reversed compared to the true effect of X on Y . The goal of multivariate regression is to “hold Z constant” so the coefficient on X reflects X ’s own contribution.

1.2 What Does “Controlling For” Actually Mean?

When statisticians say a regression “controls for” a variable Z , they mean something concrete: the coefficient on X is the relationship between X and Y *among observations that have the same value of Z* . Instead of comparing all units together, we compare like with like.

Suppose we ask whether education predicts support for immigration. A simple comparison shows that college graduates are more supportive than non-graduates. But college graduates are also disproportionately Democrats, and Democrats are more supportive of immigration for reasons unrelated to schooling. Maybe the education gap is really a party gap in disguise.

To check, we can subset the data:

- Among Democrats only, is there still an education gap?
- Among Republicans only, is there still an education gap?

If the gap survives within both parties, education is doing real work beyond partisanship. If the gap shrinks or vanishes within parties, the simple comparison was misleading — party was the true driver.

Multivariate regression performs this kind of comparison *across all values of Z simultaneously*, in a single equation. With one extra variable in the model, we are no longer fitting one line through the data; we are fitting a separate line for each value of Z and reporting their shared slope on X .

1.3 Why Not Just Subset the Data?

If subsetting on Z reveals the true effect of X , why not always do that instead of writing a multivariate regression?

There are two practical reasons. First, Z is often continuous. If we want to hold age constant, we cannot split the sample into age = 31, age = 32, age = 33, and so on — most subgroups will have only a handful of people, and many will be empty. Second, real political analyses usually require controlling for several variables at once. With four controls — party, income, age, and region — we could need hundreds of subgroups to compare like with like, and most of them would contain zero or one observation.

Regression is, in effect, automatic subsetting. It uses every observation at once, compares units with similar Z values, and produces a single, clean estimate of X ’s effect. It handles continuous

and categorical controls in the same framework, and it scales naturally to many controls without exploding the number of subgroups.

The takeaway: A multivariate regression coefficient on X is the average X–Y relationship among observations that are identical on all the other variables in the model. It is what you would find if you could perfectly subset the data and compare like with like.

1.4 Omitted Variable Bias

When we leave an important confounder out of the model, the coefficient on X is biased: it does not equal the true effect of X on Y. This is called omitted variable bias, and its direction follows a simple rule. If Z has a positive effect on Y and is positively correlated with X, leaving Z out makes the coefficient on X biased upward. If Z has a positive effect on Y but is negatively correlated with X, leaving Z out biases the coefficient on X downward.

A classic political example: the simple correlation between campaign spending and vote share is strongly positive. But incumbents tend to spend more *and* tend to receive more votes for reasons unrelated to spending (name recognition, district advantages, fundraising networks). A simple regression of vote share on spending will overstate the effect of an extra dollar, because some of what looks like a “spending effect” is really an “incumbency effect” hidden inside the residuals. Including incumbency as a second predictor pulls that confounding effect out and lets the spending coefficient reflect spending alone.

```
# Simple model: vote share on spending alone
simple_model <- lm(vote_share ~ spending, data = campaign_data)
tidy(simple_model)

# A tibble: 2 x 5
  term          estimate std.error statistic    p.value
<chr>         <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)   26.4       1.08      24.5 9.52e-62
2 spending       8.06       0.418     19.3 2.35e-47

# Multivariate model: spending plus incumbency
multi_model <- lm(vote_share ~ spending + incumbent, data = campaign_data)
tidy(multi_model)

# A tibble: 3 x 5
  term          estimate std.error statistic    p.value
<chr>         <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)   29.8       0.646     46.1 1.86e-107
2 spending       3.96       0.318     12.5 1.17e- 26
3 incumbent     15.1       0.760     19.9 5.00e- 49
```

What this code does: We fit two models predicting vote share. The simple model uses only spending and produces a large, misleading coefficient because incumbents spend more *and* win more. The multivariate model adds incumbency status. The spending coefficient drops to its true value: incumbents simply start with a higher baseline, and once we account for that, spending’s effect is what it always was. The change in the spending coefficient between models is omitted variable bias in action.

1.5 Parallel Lines: One Slope, Shifted Intercepts

The simplest multivariate model with two predictors has the form:

$$\hat{Y} = \alpha + \beta_1 X_1 + \beta_2 X_2 \quad (1)$$

There is one intercept and *two* slopes. Each slope is the effect of that variable holding the other constant. Geometrically, instead of fitting one line through the data, we are fitting parallel lines — one for each value of the second variable — that share the same slope on X_1 but are shifted up or down by β_2 .

For the campaign example, the model says: $\widehat{Vote} = \alpha + \beta_1 \cdot Spending + \beta_2 \cdot Incumbent$. Plugging in $Incumbent = 0$ gives the line for challengers; plugging in $Incumbent = 1$ gives the line for incumbents, shifted up by β_2 . The slope on spending is the same in both lines: that is the assumption of “no interaction.” The second half of this chapter relaxes that assumption.

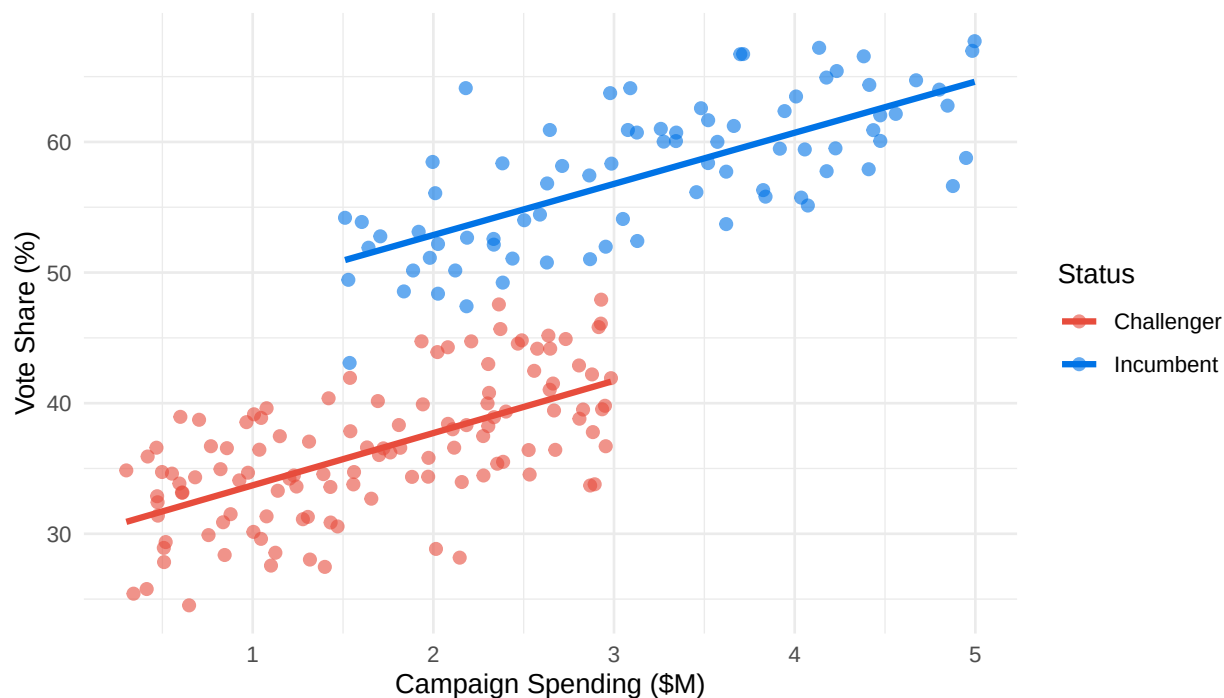


Figure 1: Figure 5.3: Multivariate regression fits parallel lines — same slope on spending, shifted intercepts by incumbency.

1.6 Interpreting Multivariate Coefficients

Each coefficient in a multivariate model tells you the effect of that variable among observations that are identical on all the other variables in the model. For the campaign example:

- The *spending* coefficient is the average change in vote share per extra \$1M spent, *among candidates with the same incumbency status*.

- The *incumbent* coefficient is the average vote-share advantage incumbents enjoy, *among candidates with the same level of spending*.

This phrasing — “holding the other variables constant” or “among observations with the same value of Z” — is the most important habit you can build in regression interpretation. A multivariate coefficient is not a marginal effect in the abstract; it is a comparison within a specific slice of the data.

1.7 When Controls Help and When They Hurt

A natural reaction to omitted variable bias is to throw everything you can measure into the model. This is a mistake. Controls do not automatically improve a regression. Whether to include a variable depends on what role it plays in the causal story.

- **Include a confounder.** A variable that affects both X and Y should generally be controlled for. Leaving it out biases the estimate of X.
- **Do not include a mediator.** A variable that is a *consequence* of X and a *cause* of Y is part of how X affects Y. Controlling for it absorbs the very effect you are trying to estimate. If education raises income and income improves health, controlling for income while estimating the effect of education on health throws away most of the story.
- **Do not include post-treatment variables.** Anything measured after X happened is potentially a mediator or a downstream consequence. Treating it as a control can introduce bias rather than remove it.
- **Do not include irrelevant variables.** Variables unrelated to X or Y add no useful information; they just chew up degrees of freedom and inflate standard errors.

Do not control for a mediator

If you want the effect of a campaign ad on vote choice, do not control for whether the voter recalls the ad. Recall is a consequence of the ad, not a confounder. Controlling for it makes the ad look like it has no effect, because you have closed the very channel through which it operates.

1.8 A Sequential Example: Presidential Approval

A useful way to see how controls reshape coefficients is to build up a model one variable at a time. Suppose we want to predict presidential approval using a few simple covariates: the unemployment rate, an indicator for the post-9/11 era, and an indicator for the post-2008 financial crisis.

```
m1 <- lm(approval ~ unemployment, data = approval_seq)
m2 <- lm(approval ~ unemployment + post_9_11, data = approval_seq)
m3 <- lm(approval ~ unemployment + post_9_11 + post_2008, data = approval_seq)
```

```
tidy(m1)
```

```
# A tibble: 2 x 5
  term          estimate std.error statistic    p.value
<chr>         <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)    64.5       1.28      50.5 5.09e-148
2 unemployment  -2.37       0.208    -11.4 3.27e- 25
```

```
tidy(m2)
```

```
# A tibble: 3 x 5
  term          estimate std.error statistic    p.value
<chr>         <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)    61.5       1.19      51.7 1.33e-150
2 unemployment  -2.25       0.186    -12.1 1.01e- 27
3 post_9_11      5.46       0.616      8.86 7.34e- 17
```

```
tidy(m3)
```

```
# A tibble: 4 x 5
  term          estimate std.error statistic    p.value
<chr>         <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)    61.2       1.16      52.6 3.10e-152
2 unemployment  -2.21       0.182    -12.1 8.70e- 28
3 post_9_11      7.78       0.856      9.09 1.46e- 17
4 post_2008     -3.63       0.950     -3.82 1.61e-  4
```

What this code does: Each model adds one variable. Watch how the unemployment coefficient changes as we add controls. The post-9/11 dummy absorbs the rally-around-the-flag boost in approval; the post-2008 dummy absorbs the financial-crisis hit. Once both are in the model, the unemployment coefficient reflects the part of the approval–economy relationship that is not driven by those structural shifts. A coefficient that stays stable across specifications is more credible than one that swings wildly.

1.9 R-Squared in Multivariate Models

Adding a predictor will (almost) always increase the R-squared, even if the predictor is pure noise. That is because OLS has more flexibility to fit the data when it has more knobs to turn. This is why we report *adjusted* R-squared in multivariate models: it penalizes the score for each additional predictor, so adding a useless variable can actually lower it.

But neither R-squared nor adjusted R-squared tells you whether your model is correctly specified. A model with R-squared of 0.80 built entirely on confounded or mediator variables is worse than a model with R-squared of 0.20 that correctly captures the causal structure. R-squared is a measure of fit, not of truth. Theory and design — not goodness of fit — should drive which variables you include.

R-squared hunting

“R-squared hunting” is the practice of adding variables until the fit looks impressive. It is dan-

gerous: it inflates fit on your sample while making predictions on new data worse (overfitting), and it lets meaningless variables in the back door. A model with $R\text{-squared} = 0.30$ that makes theoretical sense is better than $R\text{-squared} = 0.80$ built from kitchen-sink covariates.

1.10 What Multivariate Regression Can and Cannot Do

Multivariate regression is powerful, but it is not magic. It can:

- Control for *observed* confounders.
- Estimate partial relationships — the effect of X holding other measured variables fixed.
- Test whether a bivariate relationship survives once we add plausible alternatives.

It cannot:

- Control for variables you did not measure.
- Prove causation from observational data on its own.
- Fix a bad research design or a non-representative sample.

This is the bridge between regression and causal inference. Earlier in the course we saw that experiments use randomization to balance all confounders — observed and unobserved — in expectation. Observational regression can only balance the confounders you thought of and measured. Multivariate models bring us closer to causal claims than simple regression does, but they still rest on the assumption that we have not missed an important variable. That assumption can never be fully tested. It can only be defended with theory, prior research, and design.

Tying the segments together

Simple regression (Segment A) describes how two variables move together. Multivariate regression (Segment B) is how we begin to ask why. We almost never report a simple bivariate regression as our final answer; instead, we use it as a starting point and then add the controls that theory and prior work demand. The honest version of any observational regression always carries a caveat: "controlling for the variables we measured." Causal claims require more than that — they require a design that handles the confounders we cannot see.

2 Categorical Predictors and Interactions

So far our regression predictors have been numbers: campaign spending in dollars, age in years, GDP growth in percentage points. Political reality is messier. Voters belong to parties. Respondents identify as men or women. States lie in the Northeast, South, Midwest, or West. These variables carry real information about behavior, but they are not numbers. To use them in regression we need a way to translate categories into something the `lm()` function can multiply by a

coefficient.

This chapter develops that translation. We introduce factors, the R type designed for categorical data, and the dummy variables that factors expand into behind the scenes. We then extend the model with interaction terms, which let the relationship between a predictor and an outcome depend on context. The same dollar of campaign spending may matter more in a competitive district than in a safe one; the same year of education may shift partisan attitudes more for some groups than for others. Interactions formalize these conditional patterns.

2.1 Beyond Numbers: Why Categories Matter

Many of the most interesting political variables are inherently categorical. Party identification distinguishes Democrats from Republicans from Independents. Vote choice records which candidate a respondent supported. Region tells us where a state sits in the country. Gender, race, religion, and nationality all classify people into groups rather than measure them on a continuous scale.

A naive solution would be to assign numbers arbitrarily: Republican = 1, Democrat = 2, Independent = 3. This is a mistake. Regression treats numerical predictors as having meaningful spacing. If we use these codes, the model will assume that moving from Republican to Democrat changes the outcome by the same amount as moving from Democrat to Independent. There is no reason to believe that, and the coefficients become impossible to interpret.

The correct solution is to convert each category into its own indicator. We create a variable that equals 1 when an observation belongs to a category and 0 otherwise. These indicators are called dummy variables. A regression with dummy variables compares each category to a chosen reference group, which is exactly the comparison we usually want to make.

From Categories to Dummies

For a variable with K categories we create K-1 dummies. One category is held out as the reference, and every coefficient in the model represents a difference from that reference group. The reference choice is arbitrary, but the comparisons it produces should match the substantive story you want to tell.

2.2 Factors in R

R has a dedicated data type for categorical variables called a factor. A factor stores both the underlying values and a list of allowed levels, which lets R know how to order categories and which one to use as the reference. We build factors with the `factor()` function and inspect their levels with `levels()`.

```
party <- c("Republican", "Democrat", "Independent", "Democrat", "Republican")
party_factor <- factor(party, levels = c("Independent", "Democrat", "Republican"))
levels(party_factor)
```

```
[1] "Independent" "Democrat"    "Republican"
```


The levels argument fixes both the set of allowed values and their order. The first level listed becomes the reference category. Here Independent is the reference, which means coefficients on partyDemocrat and partyRepublican will compare those parties to Independents.

When a factor enters a regression, R automatically expands it into the necessary dummy variables. We rarely create dummies by hand, but understanding what R does behind the scenes is essential for interpretation.

```
voters

# A tibble: 6 x 4
  voter_id party      democrat_dummy republican_dummy
  <int> <chr>          <dbl>          <dbl>
1     1 Republican      0              1
2     2 Democrat       1              0
3     3 Independent    0              0
4     4 Democrat       1              0
5     5 Republican      0              1
6     6 Independent    0              0
```

With Independent as the reference, two dummies are enough to recover the original three-category variable. The Democrat dummy equals 1 only for Democrats, the Republican dummy equals 1 only for Republicans, and Independents are identified by zeros on both. A third dummy for Independent would be redundant and would prevent the model from being estimated.

2.3 Interpreting Categorical Predictors

Consider survey data on presidential approval, with party affiliation as the only predictor. We fit a regression using the factor directly.

```
approval_model <- lm(approval ~ party, data = survey)
tidy(approval_model) %>% select(term, estimate)

# A tibble: 3 x 2
  term          estimate
  <chr>          <dbl>
1 (Intercept)    43.6
2 partyDemocrat  19.2
3 partyRepublican -16.9
```

The intercept of about 43.6 is the average approval rating among Independents, our reference group. The Democrat coefficient of 19.2 says Democrats approve at a rate about 19.2 points higher than Independents. The Republican coefficient of -16.9 says Republicans approve at a rate about 16.9 points lower. Every coefficient is a comparison to the reference category.

The same coefficients let us compare any two non-reference groups by subtraction. The Democrat-Republican gap is roughly 36.1 points, which we obtain by subtracting the Republican coefficient from the Democrat coefficient.

2.4 Choosing a Reference Category

The reference category is arbitrary in the sense that no model fit changes when we pick a different one. The predicted values, the residuals, and the R-squared are identical. What changes is what the coefficients describe. A wise choice of reference makes the regression table easier to read.

```
survey <- survey %>%  
  mutate(party = relevel(party, ref = "Democrat"))  
approval_model2 <- lm(approval ~ party, data = survey)  
tidy(approval_model2) %>% select(term, estimate)
```

```
# A tibble: 3 x 2  
  term          estimate  
  <chr>         <dbl>  
1 (Intercept)    62.8  
2 partyIndependent -19.2  
3 partyRepublican -36.1
```

Now Democrats serve as the reference, and the intercept reports the Democratic average directly. The Independent coefficient shows how Independents compare to Democrats, and the Republican coefficient shows the Republican-Democrat gap. The underlying party means are unchanged, but the comparisons we get for free are different.

Pick a reference that makes substantive sense. If your research question is whether Democrats and Republicans differ from Independents, leave Independents as the reference. If you want to highlight differences from the majority party, use that party. The default behavior of `factor()` is to use alphabetical order, which is rarely what you want.

Always Inspect Your Reference Category

If you do not set levels explicitly, R picks the first one alphabetically. "Autocratic" beats "Democratic," "Female" beats "Male," and "Black" beats "White" in the alphabet. Coefficients then compare to whichever category happens to come first. Use `factor(x, levels = c(...))` or `relevel()` to set the reference deliberately, and report it clearly in your write-up.

2.5 Multiple Categorical Predictors

Categorical variables combine freely with each other and with continuous variables. Suppose we model presidential approval as a function of both party affiliation and gender.

```
multi_model <- lm(approval ~ party + gender, data = survey2)  
tidy(multi_model) %>% select(term, estimate)
```

```
# A tibble: 4 x 2  
  term          estimate  
  <chr>         <dbl>  
1 (Intercept)    46.5  
2 partyDemocrat   15.6  
3 partyRepublican -18.0  
4 genderFemale    2.20
```

Each coefficient now has a conditional interpretation. The Democrat coefficient compares Democrats to Independents among respondents of the same gender. The Female coefficient compares women to men holding party constant. Multiple categorical predictors work just like multiple continuous predictors: every coefficient is a partial effect controlling for the others.

2.6 Interactions: When Effects Depend on Context

An interaction occurs when the effect of one variable depends on the value of another. The relationship between X and Y is not constant; it varies across levels of a third variable Z . Interactions are common in political science because context shapes behavior. The same policy may land differently in different states. The same economic conditions may help incumbents and hurt challengers. The same campaign ad may persuade swing voters and energize partisans.

The interaction model takes the form:

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 (X \times Z) + \epsilon \quad (2)$$

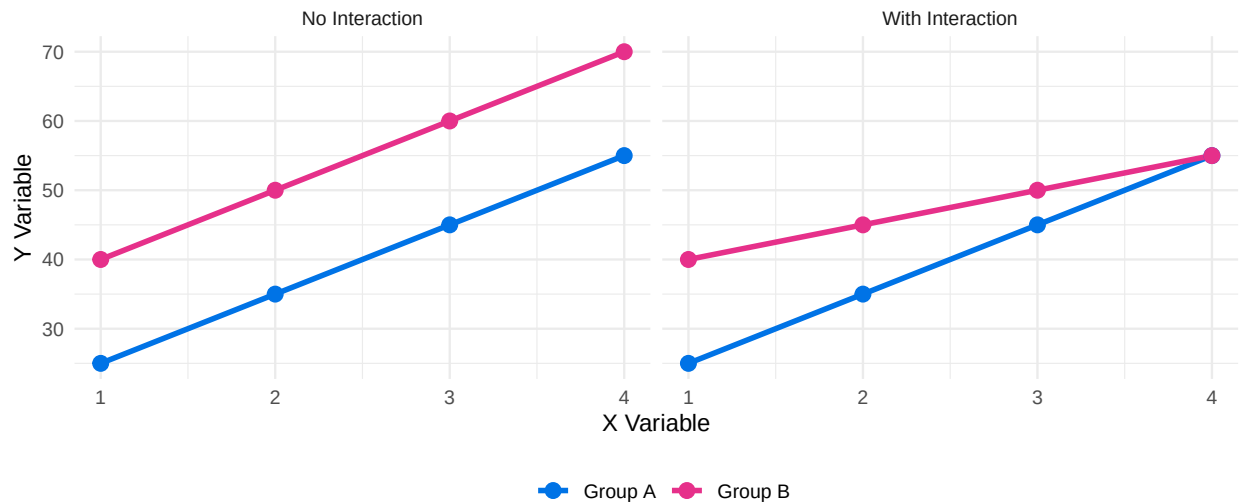
The coefficient β_3 captures how the effect of X changes as Z increases. When Z equals zero, the effect of X is simply β_1 . When Z equals one, the effect of X becomes $\beta_1 + \beta_3$. The main effect coefficients are now conditional: they describe what happens when the other variable equals zero, not the average across all observations.

Main Effects vs. Conditional Effects

Without an interaction, a coefficient is the same for everyone. With an interaction, the main effect of X is only the X effect when $Z = 0$. To get the X effect at $Z = 1$, you add the interaction coefficient. To get the X effect at any other value of Z , plug Z into the formula $\beta_1 + \beta_3 \cdot Z$.

2.7 Visualizing Interactions: Parallel vs. Non-Parallel Lines

The cleanest visual test for an interaction is to plot the relationship between X and Y separately for different values of Z and look at the slopes. Parallel lines mean no interaction: the effect of X is the same across groups, even if the groups start at different baselines. Non-parallel lines mean an interaction: the effect of X depends on which group you are in.



The left panel shows two groups with the same slope. Both groups gain ten units of Y for every one-unit increase in X. The only difference between groups is where they start, which a simple control variable handles cleanly. The right panel shows two groups with different slopes. Group A gains ten units of Y per unit of X; Group B gains only five. The effect of X depends on which group you belong to, which is exactly what an interaction term captures.

2.8 Implementing Interactions in R

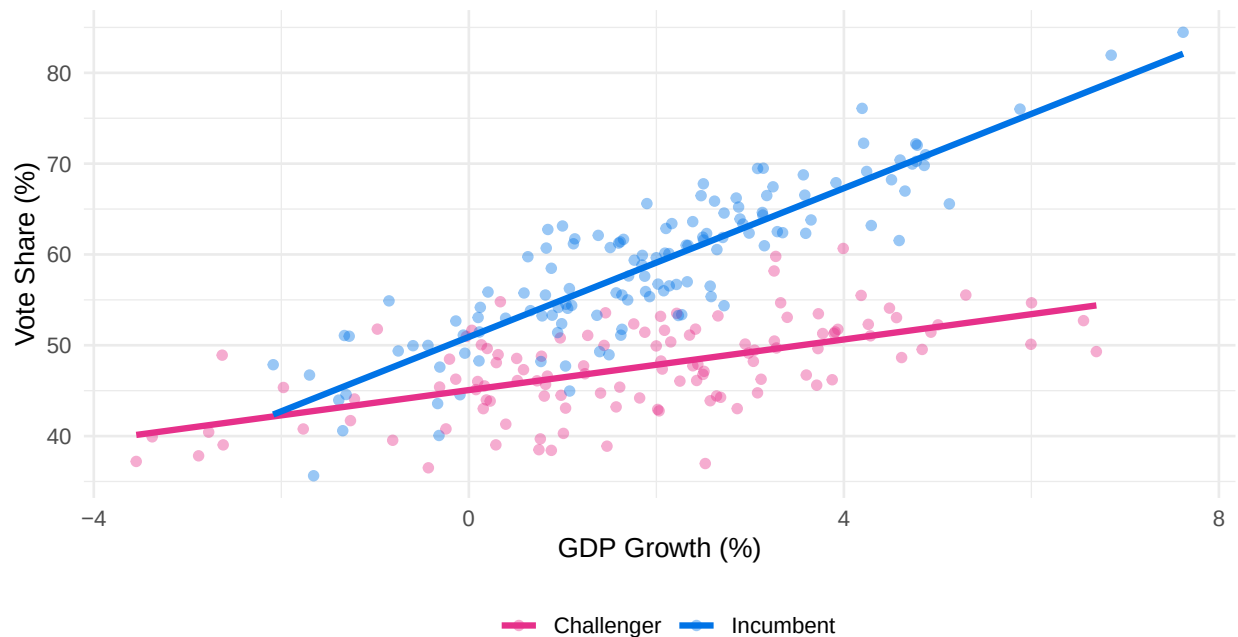
R offers a compact notation for interactions. The asterisk in a formula creates both main effects and the interaction term automatically. Writing $y \sim x * z$ is equivalent to writing $y \sim x + z + x:z$; the colon by itself produces just the product term.

Consider whether economic conditions affect vote share differently for incumbents and challengers. Voters typically reward or punish whoever they hold responsible for the economy, which is the incumbent. Challengers ride a smaller fraction of the same wave.

```
econ_model <- lm(vote_share ~ incumbent * gdp_growth, data = elections)
tidy(econ_model) %>% select(term, estimate)
```

```
# A tibble: 4 x 2
  term                estimate
  <chr>              <dbl>
1 (Intercept)        45.1
2 incumbent           5.85
3 gdp_growth          1.39
4 incumbent:gdp_growth 2.70
```

The `gdp_growth` coefficient of about 1.4 is the GDP effect for challengers (`incumbent = 0`). The interaction coefficient of about 2.7 tells us how much more GDP matters for incumbents. Adding the two gives the GDP effect for incumbents: $1.4 + 2.7 = 4.1$. A one-point increase in GDP growth lifts an incumbent's vote share by roughly 4.1 points, more than three times the effect for a challenger.



The figure tells the same story visually. The challenger line has a gentle upward slope; the incumbent line is much steeper. Voters hold incumbents accountable for the economy. Challengers ride the wave only weakly, because they did not create the conditions voters are evaluating.

2.9 Another Example: Education and Partisanship

Education affects political attitudes, but the effect is not the same across the spectrum. We might theorize that education increases trust in mainstream media for some groups and decreases it for others, because more education exposes people to a broader range of sources and to critiques of those sources.

```
trust_model <- lm(trust_media ~ education * conservative, data = partisans)
tidy(trust_model) %>% select(term, estimate)
```

```
# A tibble: 4 x 2
  term                estimate
  <chr>              <dbl>
1 (Intercept)        2.84
2 education           0.871
3 conservative       -0.254
4 education:conservative -0.606
```

The education coefficient of 0.87 is the education effect for liberals (conservative = 0). The interaction of -0.61 shows how that effect changes for conservatives. The education effect for conservatives is $0.87 + (-0.61) = 0.27$. Education raises media trust for liberals but has roughly no effect for conservatives. The two groups diverge as education rises rather than moving in parallel.

2.10 When to Use Interactions

The decision to include an interaction should be guided by theory, not by fishing for significant terms. Good reasons to include an interaction include:

- The theory predicts that an effect should differ across groups (women and men, democracies and autocracies, competitive and safe districts).
- Previous research has documented heterogeneous treatment effects.
- The substantive question itself is conditional (“does X work better in some places than others?”).
- The policy implications change with the answer.

Poor reasons to include an interaction include adding complexity until something becomes statistically significant, including all possible two-way interactions out of caution, and using an interaction when a simple control would answer the question.

The Wrong Choice Can Mislead

Using only a control when an interaction exists hides important heterogeneity. The “average” effect may not apply to anyone. Using an interaction when a simple control would do wastes statistical power and complicates interpretation. Let theory guide the choice: does Z just shift the baseline, or does Z change the relationship?

2.11 Calculating Conditional Effects

Once we have an interaction in the model, we typically want to report the effect of X at meaningful values of Z. The formula is straightforward.

$$\text{Effect of X when Z = z} = \beta_1 + \beta_3 \times z \quad (3)$$

```
coefs <- coef(econ_model)
effect_challenger <- coefs["gdp_growth"]
effect_incumbent <- coefs["gdp_growth"] + coefs["incumbent:gdp_growth"]

cat("GDP effect for challengers:", round(effect_challenger, 2), "\n")
```

GDP effect for challengers: 1.39

```
cat("GDP effect for incumbents:", round(effect_incumbent, 2), "\n")
```

GDP effect for incumbents: 4.09

For a binary moderator the calculation is simple: use the main effect at $Z = 0$ and add the interaction at $Z = 1$. For a continuous moderator you would plug in specific values (the mean, one standard deviation above and below, the 10th and 90th percentiles) and report the effect of X at each. And if the interaction coefficient turns out to be near zero, that is a real finding, not a failure: it says the slopes barely differ and the simpler additive model was adequate.

2.12 A Worked Example: Region as a Factor

Putting the pieces together, consider a final example where we model Democratic vote margin across states as a function of region. Region is a four-category variable (Northeast, South, Midwest, West), so the factor will expand into three dummies once we choose a reference.

```
region_model <- lm(dem_margin ~ region, data = states)
tidy(region_model) %>% select(term, estimate)
```

```
# A tibble: 4 x 2
  term          estimate
  <chr>         <dbl>
1 (Intercept)    -8.57
2 regionMidwest    9.59
3 regionNortheast 19.9
4 regionWest      14.3
```

With South as the reference, the intercept of about -8.6 is the Southern average (negative because the South leans Republican). The Midwest coefficient of 9.6 means Midwestern states average about 9.6 points more Democratic than Southern states. The Northeast coefficient of 19.9 shows the largest shift from the Southern baseline. To compare Northeast to West, we subtract: $19.9 - (14.3) = 5.6$ points.

The same model could be fit with a different reference. If we wanted to emphasize how regions differ from the Northeast — a common reference in studies of the South's distinct political development — we would relevel to put Northeast first.

3 Using AI for This Material

Where AI Helps This Week

- Writing correct `lm( )` syntax for a stated specification, including interactions and factor variables.
- Explaining the sign and rough magnitude of a coefficient given the units of the variables.
- Translating a fitted regression equation into a one-sentence interpretation suitable for a paper.

Where AI Gets This Wrong This Week

- Deciding which controls to include is a theory question: adding "all available controls" is wrong, omitting confounders is wrong, and including post-treatment variables is wrong — AI cannot tell which is which in your model.
- It cannot diagnose Simpson's paradox in your data without seeing the subgroup means.

- It cannot tell you whether a continuous coefficient is “large” without knowing the variable’s units and the substantive range you care about.

A prompt that works for this week:

Explain in two sentences what the coefficient on `spending_thousands` means in the model `lm(vote_share ~ spending_thousands + incumbent, data = races)`.

A prompt that misleads:

Which control variables should I include in my regression of turnout on campaign contact?

Why it misleads: control selection is a theory and design question — what is a confounder versus a mediator versus a collider in *your* causal story — and AI will produce a confident kitchen-sink list that mixes valid controls with bias-inducing ones.

Where AI Helps This Week

- Writing `factor()` calls with explicit levels and a chosen reference category so contrasts come out in the order you want.
- Producing `ggplot2` code for a specified geometry (`geom_col`, `geom_point`, `geom_density`) given a stated mapping of variables to aesthetics.
- Suggesting a colorblind-safe palette and helping you label legends, axes, and titles consistently.

Where AI Gets This Wrong This Week

- Deciding whether to model an interaction or just include both variables additively is a theory question about whether one variable’s effect depends on the other in *your* setting.
- Picking the right chart type requires seeing the data’s shape and knowing what comparison your research question emphasizes — AI defaults to a bar chart whether or not it suits the claim.
- It will not flag when R has silently chosen an alphabetical reference category that produces nonsensical contrasts (e.g. “Independent” as the baseline party).

A prompt that works for this week:

Write `ggplot2` code for a grouped bar chart showing mean approval by party, where `party` is a factor with levels Democrat, Republican, Independent in

that order, and the bars are colored by party.

A prompt that misleads:

What is the best chart to show my findings?

Why it misleads: chart choice depends on the comparison you want the reader to make (levels vs. change vs. distribution) and on the structure of your data, neither of which AI can see — it will default to a bar chart whether or not that suits the claim.

4 Common Mistakes and How to Avoid Them

Mistake 1: Interpreting Regression as Causation

A statistically significant regression coefficient shows that X and Y are associated, possibly after controlling for some other variables. It does not prove that X causes Y. Causation requires a research design that handles unobserved confounders. Always distinguish association from causation, especially in observational data.

Mistake 2: Extrapolating Beyond the Data

The regression line describes the relationship within the range of X you observed. Predictions for X values far outside this range are unreliable — the linear approximation may not hold. If unemployment in your sample ranges from 3.5% to 9.5%, do not use the model to predict approval at 20% unemployment.

Mistake 3: Controlling for Mediators or Post-Treatment Variables

Adding “more controls” is not always better. Variables that are consequences of X — or that were measured after X happened — can introduce bias rather than remove it. Think carefully about whether a candidate control variable is a common cause or a downstream effect.

Mistake 4: Dismissing Low R-Squared

In observational political research, R-squared values of 0.10 to 0.30 are common and can still reveal important relationships. A statistically significant slope with modest R-squared tells you that X is reliably associated with Y, even if X is far from the only thing that matters.

Letting R Pick the Reference Category

If you do not set the levels explicitly, R uses alphabetical order. Your coefficients may then compare to a category you did not intend, such as "Autocratic" rather than "Democratic" or "Black" rather than "White." Always set factor levels deliberately and confirm the reference in your write-up.

Misreading Main Effects in an Interaction Model

When an interaction is present, the main effect of X is the X effect when $Z = 0$. It is not the average X effect across the sample. Conservatives' effect of education, in our earlier example, is not the education coefficient; it is the education coefficient plus the interaction coefficient.

Fishing for Significant Interactions

Adding every possible two-way interaction until something is significant produces findings that will not replicate. Pre-specify the interactions your theory predicts and report them all, including the ones that are not significant.

Stopping at the Regression Table

A regression with categorical predictors and interactions is hard to read in coefficient form. Always produce at least one figure — predicted values, group means, or interaction plots — to make the model's substantive content visible to your reader.

5 R Functions Reference

Function	Purpose	Example
<code>lm()</code>	Fit linear model	<code>lm(y ~ x, data)</code>
<code>lm()</code> (multi)	Add controls	<code>lm(y ~ x + z, data)</code>
<code>tidy()</code>	Clean coefficient table	<code>tidy(model)</code>
<code>glance()</code>	Model-level statistics	<code>glance(model)</code>
<code>augment()</code>	Add fitted values & residuals	<code>augment(model)</code>
<code>predict()</code>	Make predictions	<code>predict(model, newdata)</code>
<code>coef()</code>	Extract coefficients	<code>coef(model)</code>
<code>confint()</code>	Confidence intervals	<code>confint(model)</code>

Table 1: Key R Functions for Linear and Multivariate Regression

Task	Code
Create a factor	<code>factor(x, levels = c("ref", "b", "c"))</code>
List factor levels	<code>levels(x)</code>
Change reference category	<code>relevel(factor(x), ref = "new_ref")</code>
Regression with factor	<code>lm(y ~ party, data = df)</code>
Multiple categorical predictors	<code>lm(y ~ party + gender, data = df)</code>
Interaction term	<code>lm(y ~ x1 * x2, data = df)</code>
Interaction only	<code>lm(y ~ x1 + x2 + x1:x2, data = df)</code>
Tidy coefficients	<code>tidy(model)</code> from broom
Start a plot	<code>ggplot(data) + aes(x = ..., y = ...)</code>
Line plot for time series	<code>geom_line()</code>
Scatter plot for relationships	<code>geom_point()</code>
Regression line on a scatter	<code>geom_smooth(method = "lm", se = FALSE)</code>
Box plot for group comparisons	<code>geom_boxplot()</code>
Percent-formatted axis	<code>scale_y_continuous(labels = scales::percent_format)</code>

6 Practice Problems

Question 1

A simple regression of presidential approval on the unemployment rate yields a slope of -1.8 . Interpret this number in plain English. Why might this estimate be biased relative to the true effect of unemployment on approval?

Question 2

A campaign analyst runs a regression of vote share on advertising spending and finds a slope of 0.12 (12 votes per \$1,000). When she adds incumbency status to the model, the slope drops to 0.05. What does this change tell you? Which estimate is more credible, and why?

Question 3

A colleague proposes that, to estimate the effect of education on political knowledge, the model should also include income (since income is correlated with both). Another colleague argues that income is a consequence of education, so controlling for it would be a mistake. Who is right? Why does the answer matter for the size of the estimated education coefficient?

Question 4 (Computational)

Simulate 200 observations where $Y = 5 + 2X + 3Z + \epsilon$ with X and Z positively correlated. Fit both the simple regression of Y on X and the multivariate regression of Y on X and Z .

Compare the X coefficients. Which one is closer to the true value of 2? Why?

Conceptual Questions

First, you fit a regression with party (Democrat, Independent, Republican) predicting approval, and the partyRepublican coefficient is -25. What does this tell you about the comparison between Republicans and Democrats, given that Independent is the reference category? What would you need to add or subtract to get the Republican-Democrat difference?

Second, an analyst reports an interaction between income and gender in a model of support for paid leave, with a positive interaction coefficient. Explain in plain language what this interaction means. Why is the main effect of income no longer the "average" income effect?

Third, you are preparing a single figure to communicate the result of your interaction model from Segment B. Which geometry is best suited to showing predicted values from the model across a continuous predictor for each group, and what scale helpers would you use to make the axes legible?

For Further Study

This chapter introduced multivariate regression, categorical predictors, and interaction terms. Next week we turn to sampling and statistical significance: how much our estimates would vary from sample to sample, and when a coefficient is distinguishable from chance. For deeper treatment, Wooldridge's *Introductory Econometrics* gives a rigorous account of omitted variable bias; Kellstedt and Whitten's *The Fundamentals of Political Science Research* offers an intuitive walkthrough with political examples; and Angrist and Pischke's *Mostly Harmless Econometrics* connects regression to causal inference from observational data. Advanced topics to explore include three-way interactions, ordered factors for ranked variables (education, income brackets), and the `marginaleffects` and `ggeffects` packages for computing and plotting predicted values from complex models.